

Speaker Notes

Loan Performance Intelligence Engine

Intain FinTech AI Challenge 2026 - Final Round 3

Narendra's Team | September 17, 2026

Presentation Time: 04:00 PM - 07:00 PM | Sequence #9

Total Presentation: ~10 minutes (7 min slides + 3 min live demo)

Tip: Speak slowly, pause after key numbers, make eye contact with judges.

SLIDE 1 - Title & Introduction

Timing: 30 seconds

What to Say:

"Good afternoon judges and fellow finalists. I'm Narendra, and today I'm presenting the Loan Performance Intelligence Engine - or LPIE.

LPIE is an end-to-end AI-powered mortgage surveillance platform that transforms raw loan performance tapes into actionable, explainable, and fully auditable credit intelligence - in just 40 seconds.

Let me walk you through how we built it."

Transition: "Let's start with the problem we're solving."

SLIDE 2 - The Industry Problem

Timing: 45 seconds

What to Say:

"The secondary mortgage market today faces four critical problems.

First, legacy surveillance systems rely on manual, rule-based reviews. Servicers and investors often miss 30 to 40 percent of early-stage delinquencies because the review process is too slow.

Second, existing credit models treat prepayment and default as independent events. But in reality, a loan that prepays can never default - they are competing risks that censor each other. Ignoring this creates statistical bias.

Third, there is no automated way to catch servicer data errors - things like negative balances, stale records, or impossible delinquency transitions.

And fourth, when teams use LLMs to generate reviewer notes, there are zero guarantees that the output is factual. LLMs hallucinate credit facts with no audit trail.

We built LPIE to solve all four."

Transition: "Here's our end-to-end architecture."

SLIDE 3 - Solution Architecture

Timing: 60 seconds

What to Say:

"LPIE is a 9-phase automated pipeline that runs end-to-end with a single command.

It starts with raw loan performance tapes - 34,285 monthly panel records across 3,000 loans spanning 2022 to 2024 vintages.

Phase 1 runs our Data Quality Profiler, which scores the dataset at 97.66 out of 100 - Grade A quality.

Phase 2 performs leakage-safe feature engineering. We explicitly drop 6 post-event administrative columns like `default_flag` and `loss_severity_band` before any model ever sees the data.

We then train calibrated XGBoost and LightGBM models evaluated on a strict out-of-time chronological partition - training on data before November 2023, testing on data after. No loan ID ever appears in both sets.

Downstream, we run competing-risks survival modeling, hybrid anomaly detection, TreeSHAP explainability, and finally our grounded reviewer copilot that generates zero-hallucination audit notes.

The entire pipeline executes in 40 seconds with one command: `python run_pipeline.py`."

Transition: "Let me dive into each phase, starting with data quality."

SLIDE 4 - Data Intelligence & Quality

Timing: 45 seconds

What to Say:

"For the Data Intelligence phase, we built an automated profiling engine that scores data quality across three dimensions.

Completeness scored 97.4 out of 100 - we use median imputation with explicit missingness indicator columns so we never lose statistical power.

Validity scored 96.8 out of 100 - we identified 1.63 percent of records with accounting ledger errors, such as current balances exceeding original balances.

For drift monitoring, we computed the Population Stability Index across all features. Every single feature registered a PSI below 0.04, confirming zero distribution divergence between our training and validation windows.

We also ran Pearson correlation matrices and Cramer's V for categorical associations."

Transition: "With clean, validated data, let's look at our predictive models."

SLIDE 5 - Predictive Modeling (MOST IMPORTANT)

Timing: 60 seconds

What to Say:

"This is the core of our engine - multi-horizon calibrated predictive models worth 20 rubric points.

Our headline result: 12-month Default ROC-AUC of 0.8168, which is a 12.6 percent improvement over our logistic regression baseline.

For 12-month Prepayment, we achieved 0.7119 ROC-AUC - a 23.8 percent improvement - which is notably harder because prepayment behavior is driven by external rate movements.

Three things make our approach rigorous:

First, we use a strict out-of-time chronological split. Training: January 2022 to October 2023. Validation: November 2023 to April 2024. No loan ID appears in both sets.

Second, our models use isotonic probability calibration with class-weight balancing, giving us a Brier Score of just 0.1063 - meaning our predicted probabilities are well-calibrated.

Third, we run an automated leakage audit verifying zero features exceed 0.90 correlation with any target variable."

KEY PHRASE: "Out-of-time chronological validation - no loan contamination."

Transition: "But binary classification alone is insufficient. Let me explain why."

SLIDE 6 - Competing-Risks Survival

Timing: 60 seconds

What to Say:

"Standard binary classifiers treat prepaid loans and active loans as equivalent non-default outcomes. But that's statistically incorrect.

A loan that prepays in month 6 can never default in month 12. Prepayment censors the default observation. Ignoring this creates positive survival bias.

We modeled this properly using Cumulative Incidence Functions that decompose the portfolio into three mutually exclusive terminal states:

- 12.6% of loans defaulted
- 55.7% prepaid
- 31.7% remained active and right-censored

Our Log-Rank test confirmed Prime vs Subprime segments have statistically different survival curves with a p-value of 2.83 times 10 to the negative 13.

Using Cox Proportional Hazards, we found:

- Prime borrowers have 26.9% lower default hazard (HR = 0.731)
- Every 1% rate increase raises default hazard by 57.8% (HR = 1.578)

And our Weibull AFT model outperformed constant hazard by 149.8 AIC points."

KEY PHRASE: "Prepayment censors default - you cannot model them independently."

Transition: "Beyond prediction, we also catch data anomalies."

SLIDE 7 - Anomaly Detection

Timing: 45 seconds

What to Say:

"Our anomaly engine uses a hybrid approach - combining 10 deterministic business rules with unsupervised machine learning.

The rule-based layer catches concrete violations: negative balances, balances exceeding origination amounts, impossible delinquency transitions, and stale records.

The ML layer uses an Isolation Forest for spatial density estimation - it identifies records that are statistically unusual across the full feature space.

We combine both into a composite anomaly score from 0 to 1, and generate 25 reviewer-ready diagnostic case studies - each with the specific exception type, root cause, and a prescribed remediation action.

This automates what typically takes a human reviewer hours of manual tape inspection."

Transition: "What happens when the economy turns? We simulate that next."

SLIDE 8 - Macro Stress Testing

Timing: 45 seconds

What to Say:

"Our Scenario Simulation Lab stress-tests the portfolio under three macro regimes.

Under Base conditions, the 12-month default rate is 14.83% and prepayment is 50.75%.

Under Adverse Credit - simulating a 150 basis point rate shock with credit tightening - the default rate surges to 33.91%, a 2.3x increase. Meanwhile, prepayment drops 40% to 30.53% because fewer borrowers can refinance.

Under High Prepayment - simulating a 100 basis point rate drop - the default rate falls to 9.81%, but 100% of the portfolio prepays as borrowers rush to refinance.

In our live Streamlit dashboard, judges can interactively adjust the rate shock slider and see projections update in real time."

Transition: "All of these predictions need to be explainable. Here's how."

SLIDE 9 - Explainability & Responsible AI

Timing: 45 seconds

What to Say:

"Every prediction in LPIE is fully decomposable using TreeSHAP.

We provide three layers of explainability:

Global attribution - beeswarm plots ranking the top 15 features by mean absolute SHAP value across the full validation set. This tells portfolio managers which features matter most.

Local waterfall charts - for any individual loan, we show exactly how each feature pushed the prediction up or down from the baseline. This is critical for Fair Lending compliance.

Error diagnostics - we analyze our False Positives and False Negatives to understand systematic model failures.

This makes LPIE compliant with ECOA and Regulation B Fair Lending requirements, where every adverse action must be explainable."

Transition: "Now, the final intelligence layer - our Reviewer Copilot."

SLIDE 10 - Grounded Reviewer Copilot

Timing: 45 seconds

What to Say:

"Most AI reviewer tools use large language models that can and do hallucinate credit facts. We took a fundamentally different approach.

Our copilot is a deterministic, template-driven engine - not a generative LLM. It works in three steps:

Step 1: It ingests loan attributes verified against our Data Dictionary.

Step 2: It computes local TreeSHAP attribution values for that loan's prediction.

Step 3: It generates a structured reviewer note where every single risk claim must cite a numerical SHAP value. If a feature doesn't have a significant SHAP contribution, the copilot will not mention it - period.

We also include 3 explicit rejection examples showing cases where the copilot refused to generate unsupported claims.

Every prompt and output is logged to an immutable audit file for full regulatory traceability."

KEY PHRASE: "Zero hallucination - every risk claim cites a numerical SHAP value."

Transition: "Let me show you the live product."

SLIDE 11 - Live Demo Walkthrough

Timing: 2-3 minutes

Demo Step 1 - Mission Control (30 sec):

"This is our Executive Mission Control. You can see the four KPI cards - 34,285 records, 3,000 loan cohorts, DQI of 97.66, and Default AUC of 0.8168. Below, we have interactive Plotly charts showing credit quality distribution and loan status breakdown."

Demo Step 2 - Real-Time Scoring (45 sec):

"Here I can score any loan in real time. Let me set interest rate to 8 percent, DPD to 60 days, and credit band to Subprime."

[Click Calculate]

"You can see the gauge meters - the 12-month default hazard jumped and the system automatically assigned an ESCALATE triage badge. The anomaly score is also computed instantly."

Demo Step 3 - Stress Lab (30 sec):

"Here judges can drag the interest rate shock slider to plus 200 basis points and watch the default projection update in real time."

Demo Step 4 - Copilot (45 sec):

"Let me select a loan and generate a reviewer note."

[Click Generate]

"Notice every risk statement cites a specific SHAP value. There are no generic statements. Everything is grounded in data."

SLIDE 12 - Conclusion

Timing: 30 seconds

What to Say:

"To summarize - LPIE delivers across all 8 rubric dimensions:

- 97.66 DQI - enterprise-grade data quality
- 0.8168 ROC-AUC - best-in-class default prediction with out-of-time validation
- Competing risks survival modeling with p-value below 10 to the negative 13
- 25 automated anomaly case studies with remediation prescriptions
- Interactive stress testing under 3 macro scenarios
- 100 percent explainable predictions via TreeSHAP
- Zero-hallucination grounded reviewer copilot
- And a 40-second, one-command reproducible pipeline deployed live on Streamlit Cloud

Thank you. I'm happy to take any questions."

JUDGE Q&A CHEAT SHEET

Q: "How did you prevent data leakage?"

Answer: Three safeguards: First, we explicitly drop 6 post-event administrative columns before feature engineering - `default_flag`, `prepayment_flag`, `loss_severity_band`, `last_updated_at`, `source_system`, and `document_status`. Second, we run an automated correlation audit confirming no feature exceeds 0.90 correlation with any target. Third, we use strict out-of-time chronological partitioning with zero loan ID overlap.

Q: "Why competing risks instead of logistic regression?"

Answer: Because prepayment and default are mutually exclusive absorbing states. A loan that prepays in month 6 can never default in month 12. Standard logistic regression treats prepaid loans as non-defaults, which introduces positive survival bias. Our CIF modeling correctly accounts for this competing censoring.

Q: "How do you guarantee no hallucination?"

Answer: It is not a generative LLM - it is a deterministic template engine. Every risk assertion must cite a numerical TreeSHAP value. If a feature does not have a significant SHAP contribution, the copilot physically cannot mention it. We also log every prompt and output to an immutable audit trail for regulatory compliance.

Q: "What is your model's biggest weakness?"

Answer: Our prepayment model has ROC-AUC of 0.7119 - good but not excellent. Prepayment behavior is heavily driven by external macroeconomic rate movements not fully captured in loan-level features. In production, we would integrate forward rate curve data from Treasury yields.

Q: "Can this scale to millions of loans?"

Answer: Yes. Our pipeline processes 34,000 records in 40 seconds using vectorized pandas and optimized XGBoost. The FastAPI microservice supports concurrent loan scoring, and Docker containerization enables Kubernetes deployment for enterprise-scale portfolios.

Q: "Why XGBoost instead of deep learning?"

Answer: For tabular mortgage data with 12-15 engineered features, gradient-boosted trees consistently outperform neural networks in both accuracy and calibration. More importantly, tree models are natively compatible with TreeSHAP giving exact Shapley values - critical for Fair Lending compliance where explanation fidelity is legally required.